IS4990 COMPREHENSIVE GROUP PROJECT · PROJECT PROPOSAL

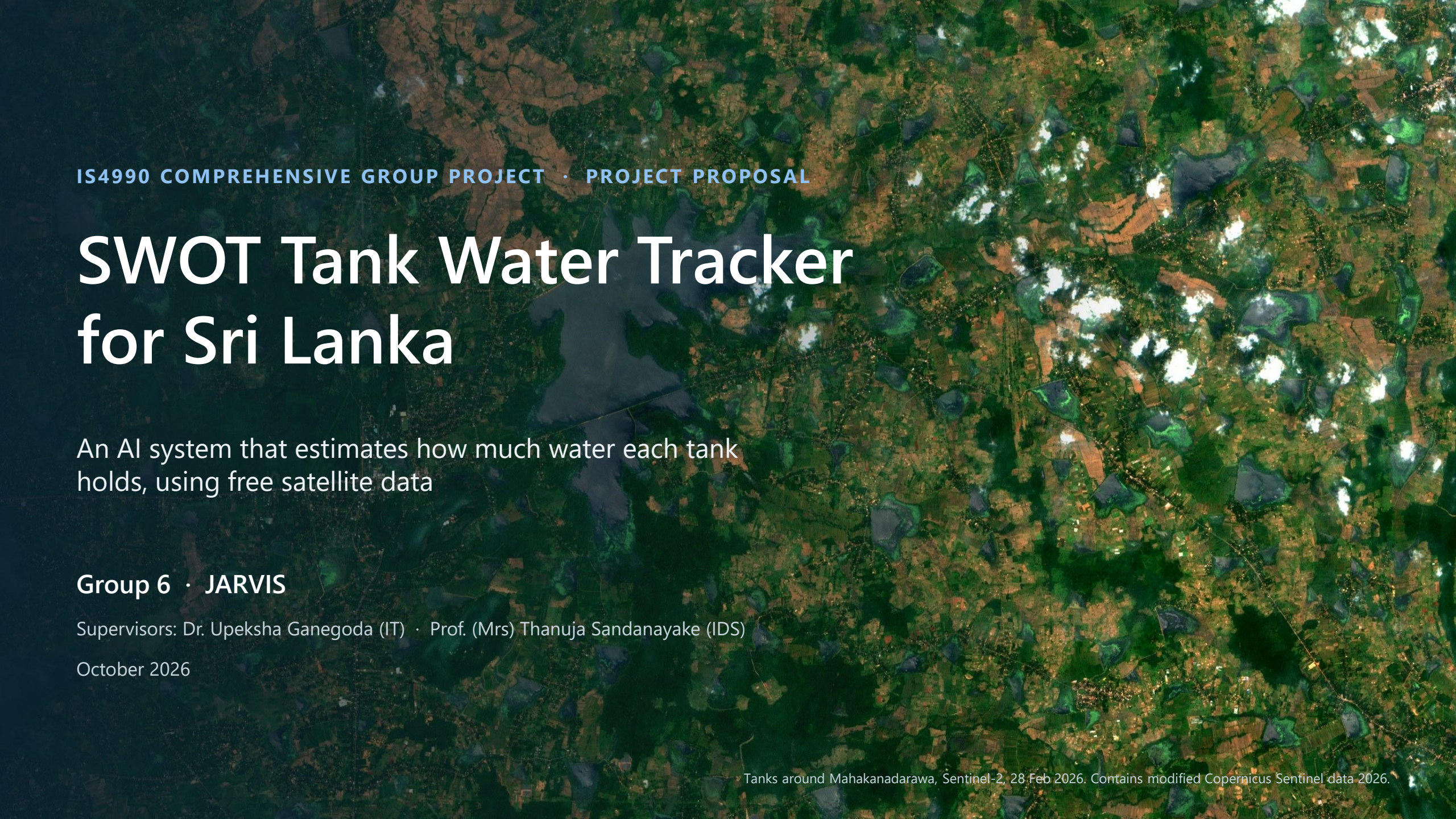
SWOT Tank Water Tracker for Sri Lanka

An AI system that estimates how much water each tank holds, using free satellite data

Group 6 · JARVIS

Supervisors: Dr. Upeksha Ganegoda (IT) · Prof. (Mrs) Thanuja Sandanayake (IDS)

October 2026



Tanks around Mahakanadarawa, Sentinel-2, 28 Feb 2026. Contains modified Copernicus Sentinel data 2026.

Why: dry-zone farming depends on tank water

~23,000

small tanks of 80 ha or less in Sri Lanka, the base of dry-zone farming

National Tanks Survey, in Lopane et al., 2025

2

seasons a year. Before each, farmers plan the crop by the water in the tank

Cultivation meetings at village tanks, Oka et al., 2015

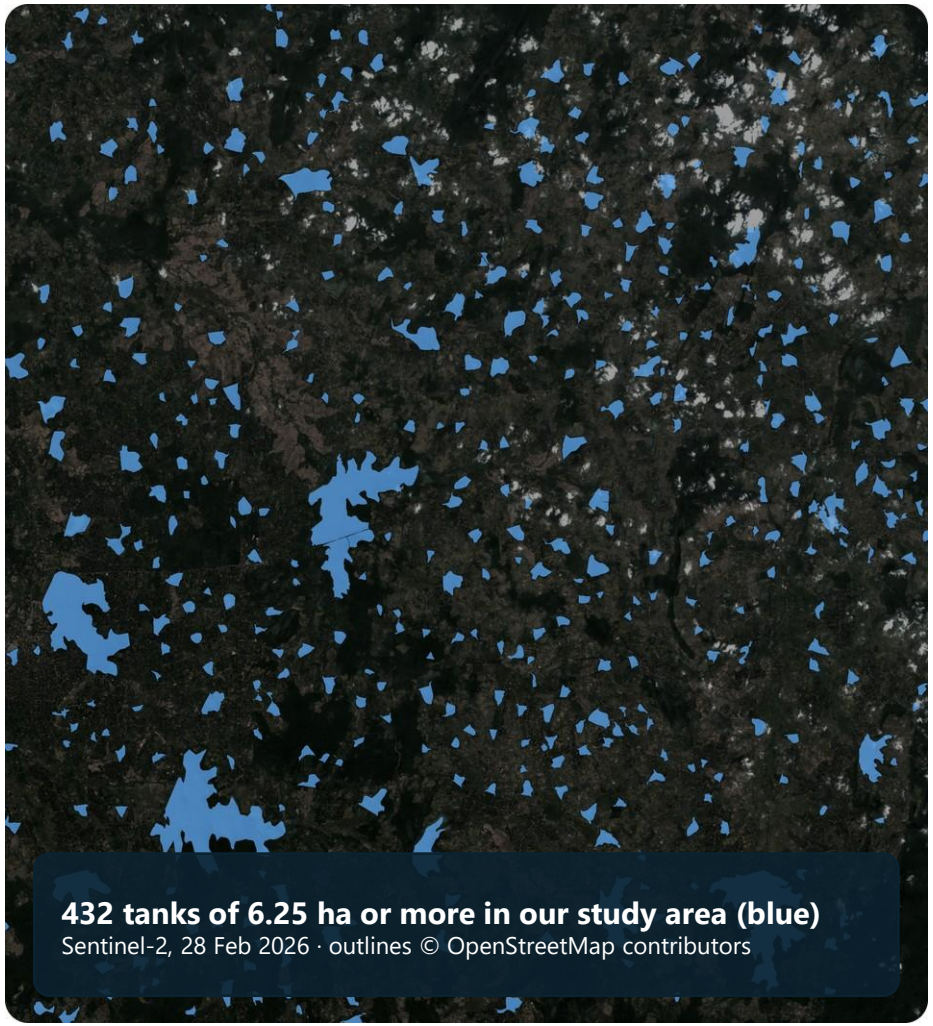
1,160

small tanks damaged by Cyclone Ditwah in November 2025

Department of Agrarian Development, December 2025

Knowing each tank's water helps farmers plan the season, and helps officers share water and prepare for floods.

But today, nobody can say how much water most tanks hold.



432 tanks of 6.25 ha or more in our study area (blue)
Sentinel-2, 28 Feb 2026 · outlines © OpenStreetMap contributors

The problem we solve

THE PROBLEM

For most Sri Lankan tanks, nobody can say how much water they hold right now, and there is no low-cost way to measure it often.

WHY IT IS STILL UNSOLVED

No gauges on village tanks

Only 118 reservoirs publish daily water levels

Irrigation Department, Reservoir Status, 25 Oct 2026

Maps show area, not volume

IFAD maps over 10,650 water bodies, but by area only

IFAD, 2026

Clouds and water plants hide the water

Only 17–22% of Sentinel-2 dates were clear over our tanks; plants cover 3 of our 8 village tanks

Our data check, October 2026

SWOT gives height, not volume

Height only, and not yet checked on Sri Lankan tanks

NASA and CNES satellite, launched December 2022

Our answer: an AI system that turns free satellite data into each tank's storage, in m^3 and % full, every 5–12 days, checked against gauges.

For farmer organisations, the Department of Agrarian Development (village tanks) and the Irrigation Department (reservoirs).

SWOT gives the height. We turn it into water

What SWOT gives

- The water height of each lake, about twice every 21 days
- Only for lakes in its prior lake list
- Mostly "suspect" readings on small village tanks
- No volume, and nothing between passes

What our project adds

- Water area from Sentinel-1 and Sentinel-2 by AI, even under cloud and water plants
- A height–area–volume curve for every tank
- Storage in m^3 and % full, every 5–12 days
- Accuracy checked against Irrigation Department gauges
- A dashboard for irrigation officers and farmers

SWOT is the ruler. We turn its readings into water in the tank.

SWOT: a satellite that measures water height

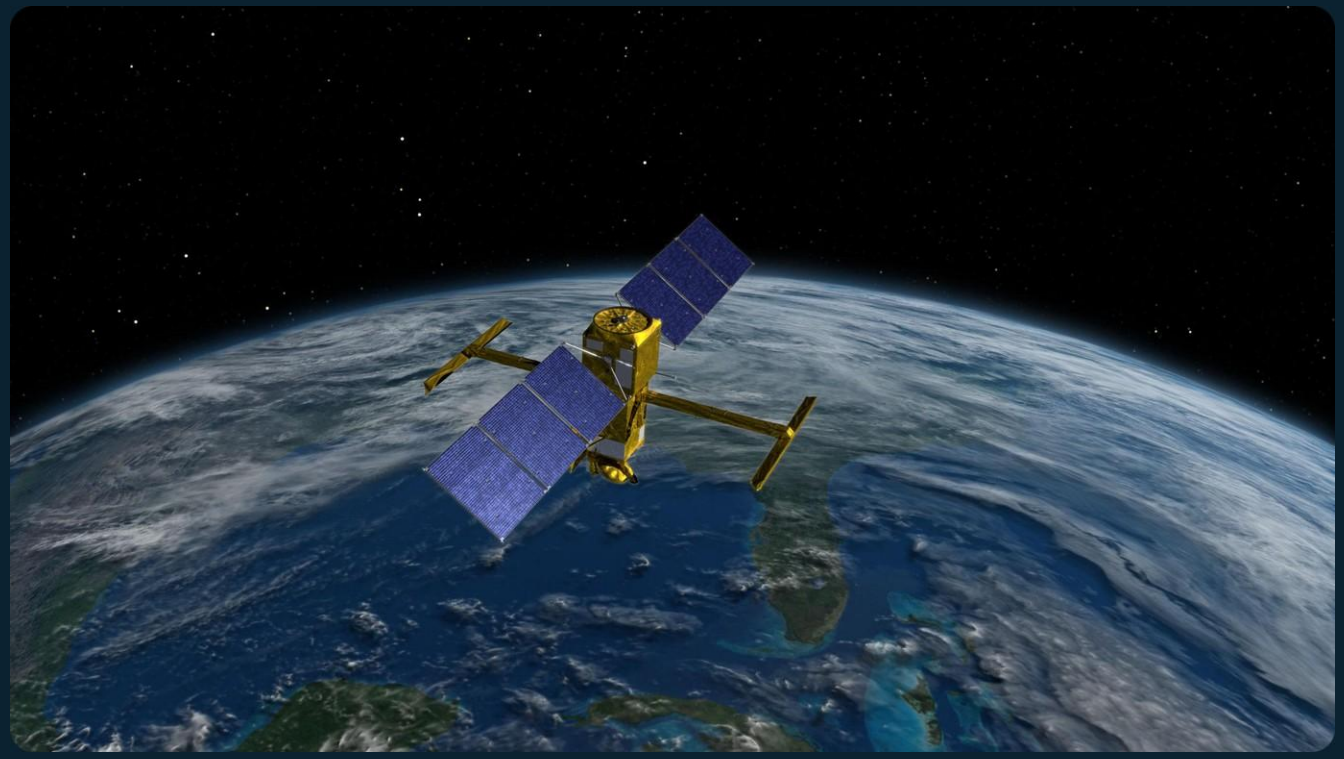


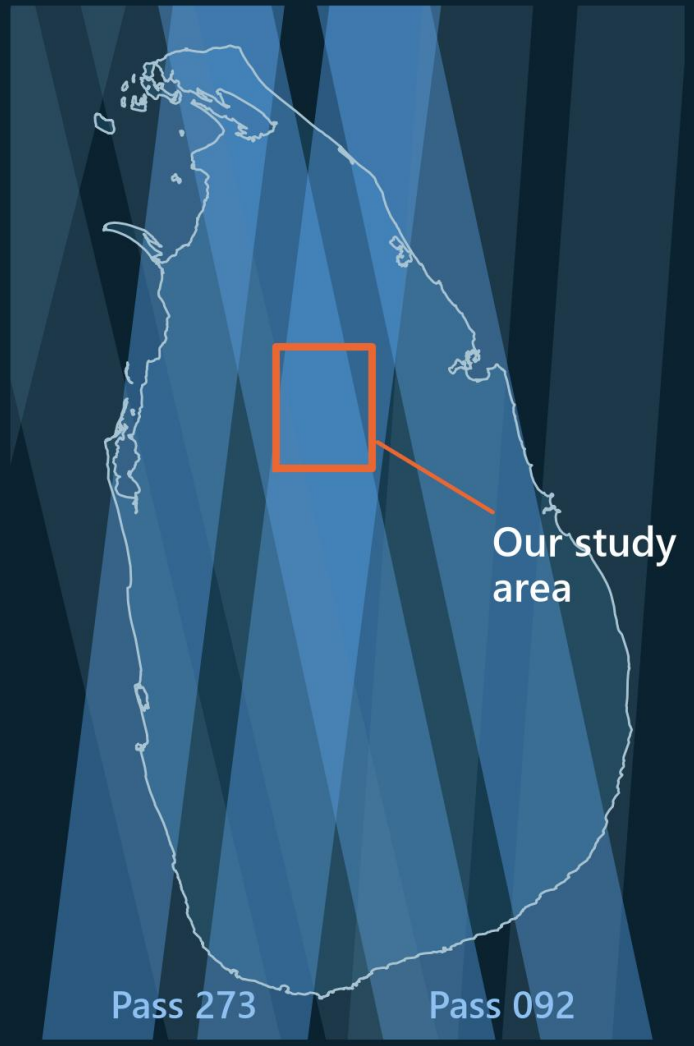
Illustration: NASA/JPL-Caltech

Dec 2022
launched by NASA and CNES (France)

120 km
wide strip measured on each pass

21 days
until the same pass comes back

2 passes
see our tanks: 273 and 092



What SWOT sees: one pass, every lake at once

198

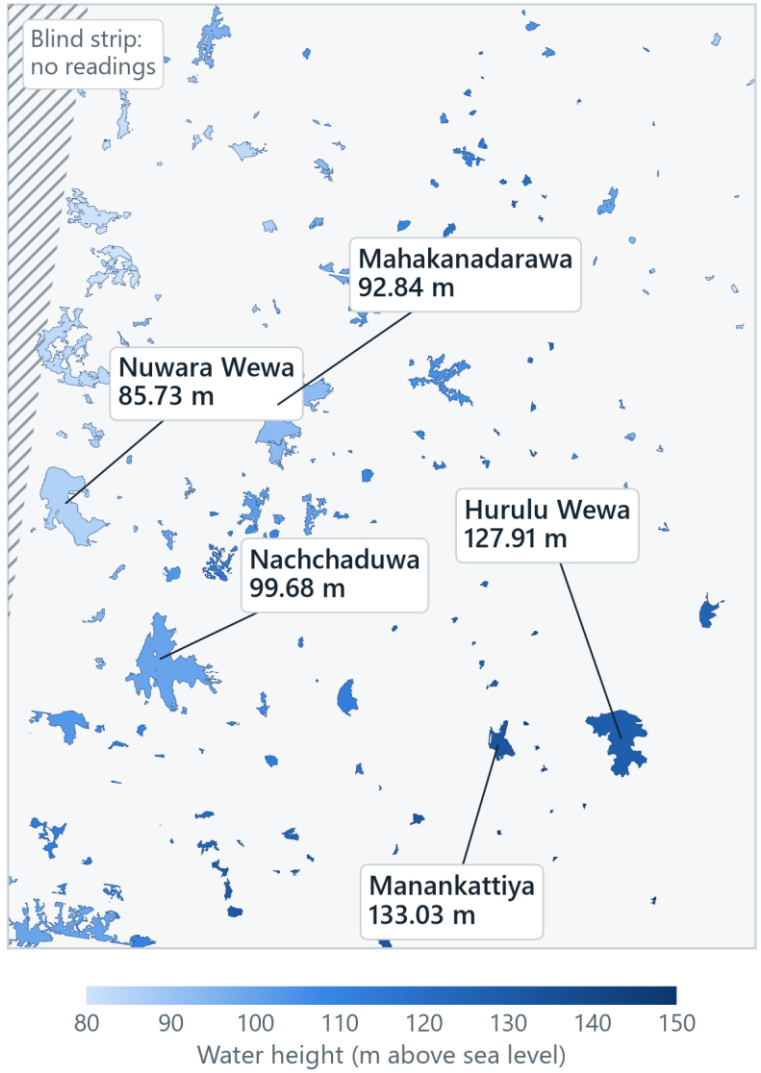
lakes measured in a single pass

SWOT pass 273 on 11 Oct 2026, in our 44 × 55 km study area

Each colour is the water height SWOT measured, in metres above sea level.

All 5 of our reservoirs were measured on this pass, for example Hurulu Wewa at 127.91 m.

SWOT cannot measure the strip right under itself, shown hatched on the map.



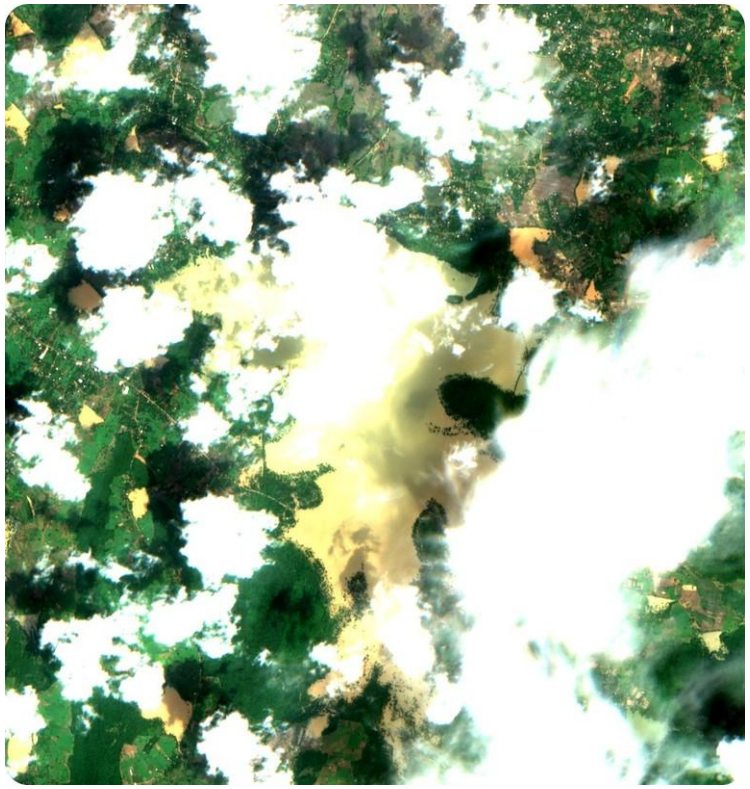
Data: NASA/JPL and CNES, SWOT Lake Single-Pass product, version D, from NASA PO.DAAC.

Clouds hide the tanks. Radar sees through them



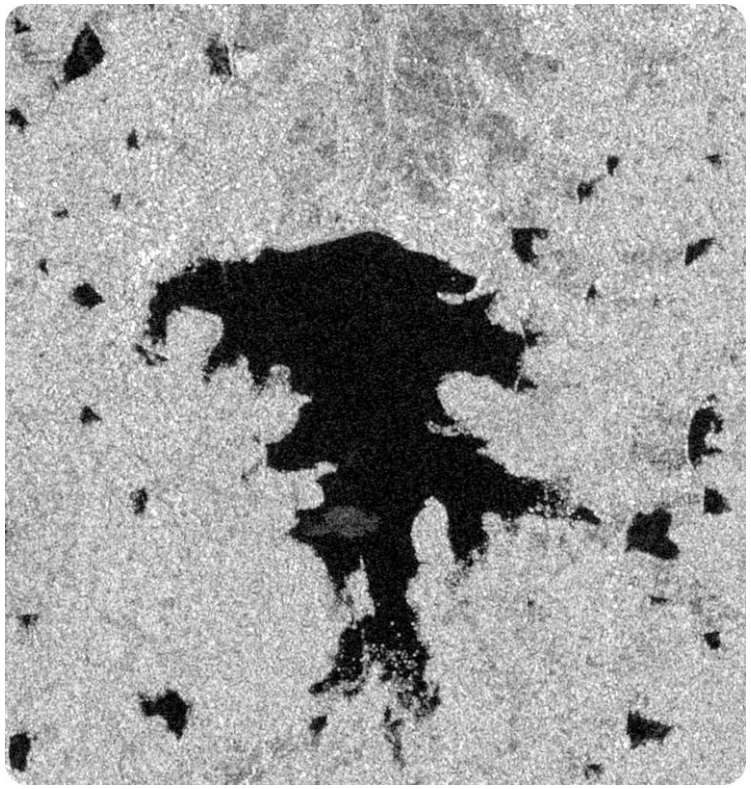
Sentinel-2, clear day

Hurulu Wewa, 28 Feb 2026 · optical, 10 m



Sentinel-2, monsoon cloud

Hurulu Wewa, 30 Nov 2025 · the tank is hidden



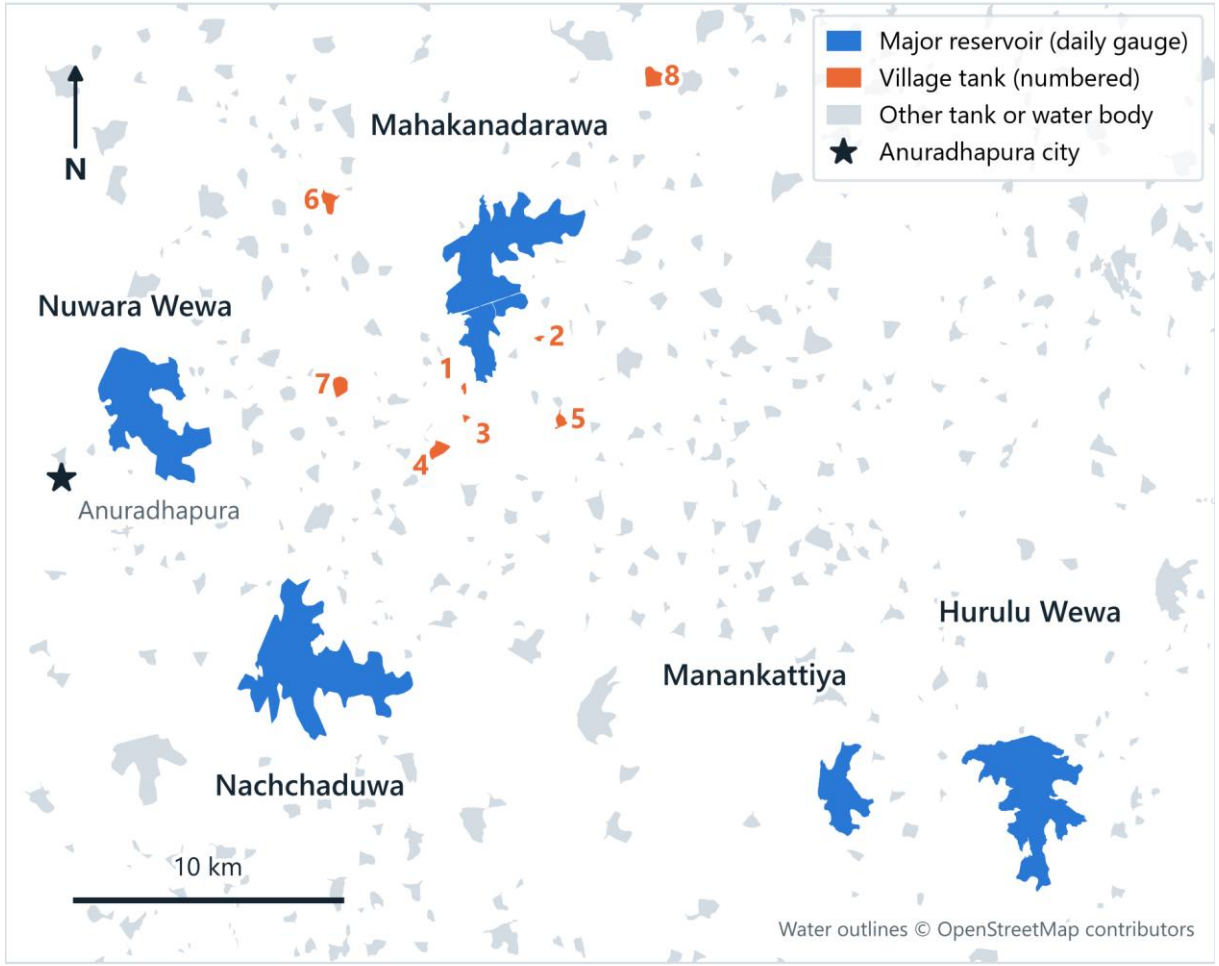
Sentinel-1 radar

Hurulu Wewa, 2 Dec 2025 · water shows dark

Only 17–22% of Sentinel-2 dates were clear over our test tanks, so radar is a core input.

Contains modified Copernicus Sentinel data 2025–2026.

Study area: 5 reservoirs and 8 village tanks



5 major reservoirs

Hurulu Wewa, Mahakanadarawa, Manankattiya, Nachchaduwa and Nuwara Wewa, all with daily gauge readings

8 village tanks

0.4–5 km from Mahakanadarawa: 2 open water, 3 mixed and 3 mostly covered by water plants

Seen by SWOT passes 273 and 092

All 13 tanks, about twice every 21 days, with 47–90 usable readings each since July 2023

Why here

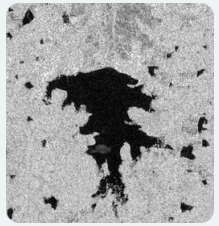
Dry-zone tank cascades (ellanga), flat land, and official gauges to check against

How it works

Sentinel-1 and Sentinel-2

Radar and optical images, every 5–12 days





AI water map

Open water, water under plants, and land



SWOT water heights

From NASA's Hydrocron service since July 2023, with poor readings removed

Height–area–volume curve

One per tank: each SWOT height paired with the AI water area of the same day

Storage every 5–12 days

Each new water map becomes a volume, shown on a dashboard

Checked against Irrigation Department gauges and official storage figures

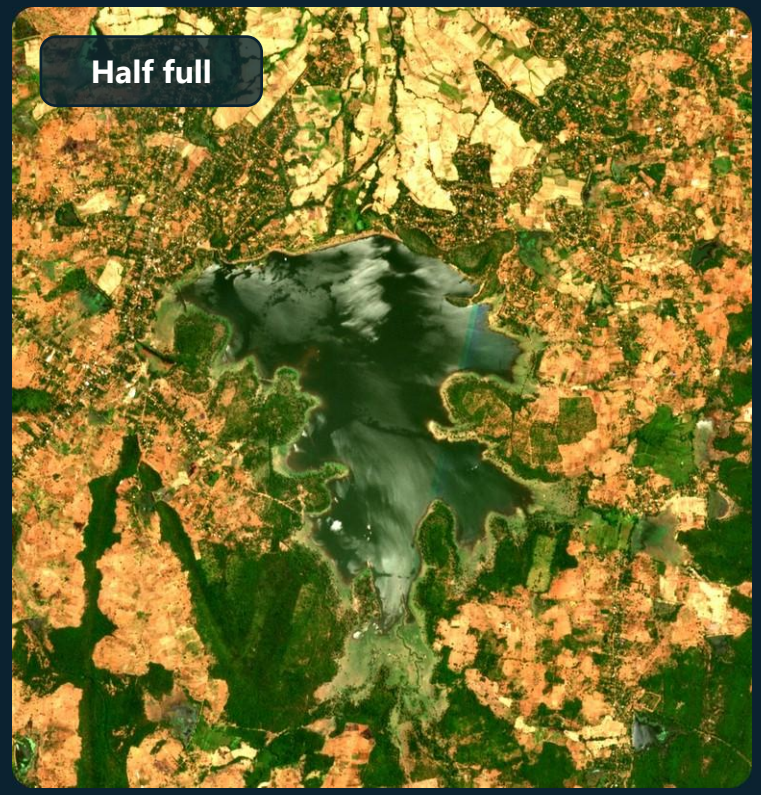
Our method in three pictures: Hurulu Wewa



127.59 m SWOT height

6.1 km² of water

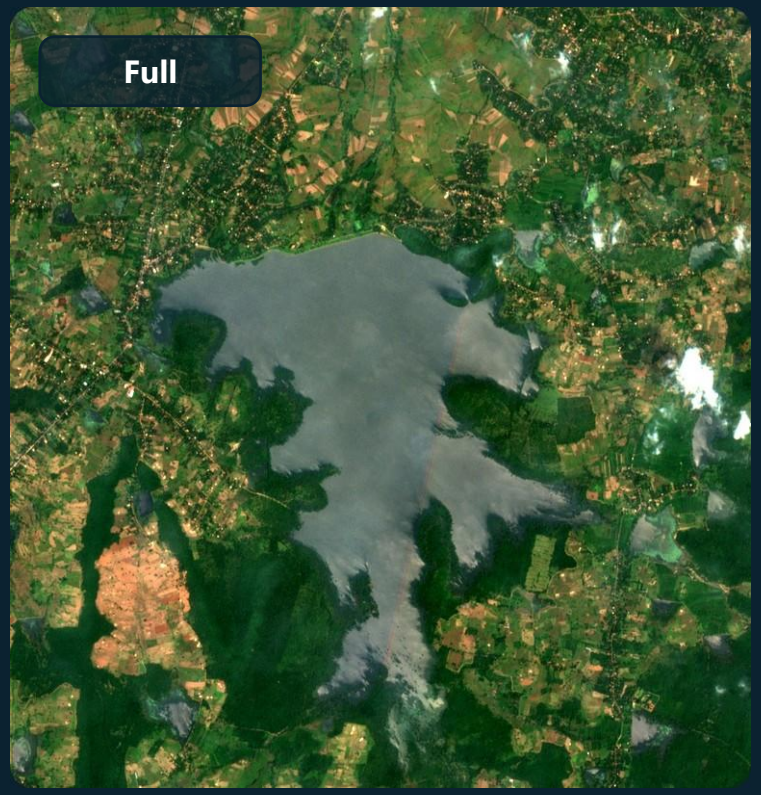
Sentinel-2 29 Jul 2023 · SWOT 30 Jul 2023



129.50 m SWOT height

8.3 km² of water

Sentinel-2 and SWOT, 30 Jul 2025



132.35 m SWOT height

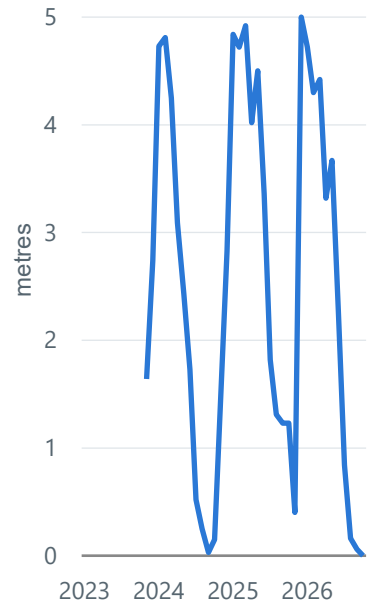
10.8 km² of water

Sentinel-2 28 Feb 2026 · SWOT 10 Mar 2026

Three years of SWOT water levels

Monthly average of best-quality readings, in metres above each reservoir's lowest month.
The sharp rise at the end of 2025 is Cyclone Ditwah.

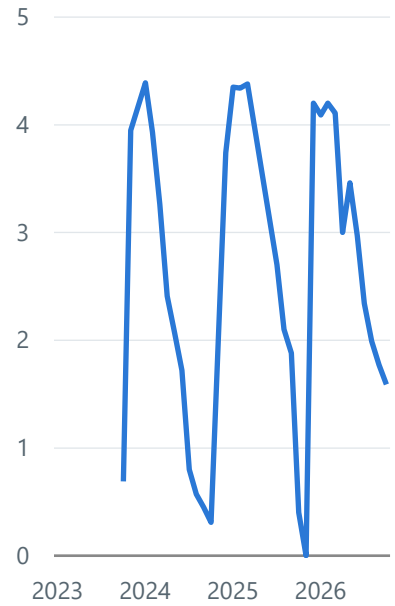
Hurulu Wewa



71 best-quality readings

rose and fell up to 5.0 m

Mahakanadarawa



63 best-quality readings

rose and fell up to 4.4 m

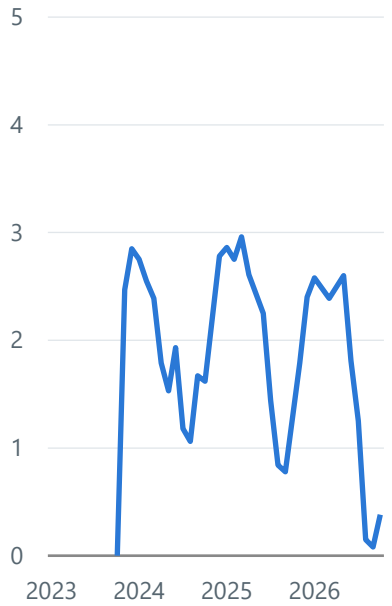
Manankattiya



57 best-quality readings

rose and fell up to 3.7 m

Nachchaduwa



55 best-quality readings

rose and fell up to 3.0 m

Nuwara Wewa



53 best-quality readings

rose and fell up to 4.3 m

Source: NASA PO.DAAC Hydrocron, SWOT lake product version D, July 2023 to October 2026. Gaps are months without a best-quality reading.

We tested the data before proposing

Five quick checks in October 2026, using the real SWOT, Sentinel-2 and gauge data.

1 · LAKE IDS

86%

of Irrigation Department tanks in our area have a SWOT lake ID (43 of 50); 58% of all mapped tanks

RISK

2 · HEIGHTS

26,641

SWOT readings on 400 lakes since July 2023; village tanks get mostly "suspect" readings

RISK

3 · GAUGE CHECK

40–64 cm

a steady gap between SWOT and the gauges at all 5 reservoirs; one fixed correction each

PASS

4 · WATER AREA

17–22%

of Sentinel-2 dates clear; water plants hide the water in 3 village tanks

RISK

5 · TRAINING DATA

2

labelled water datasets ready to use: Sen1Floods11 and S1S2-Water

PASS

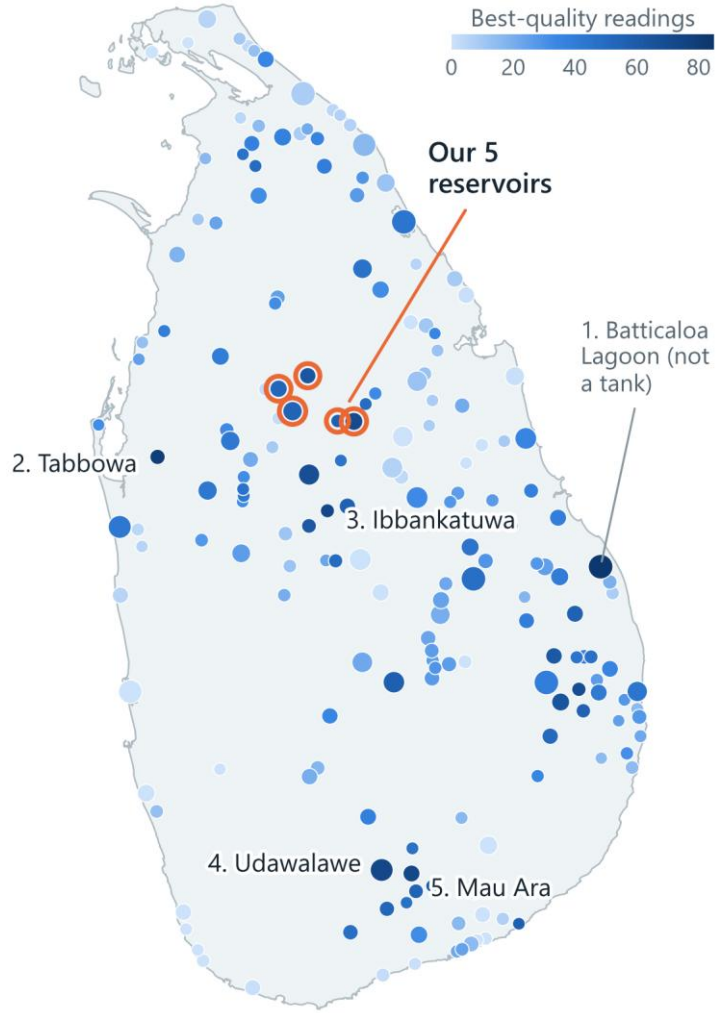
Verdict: GO, with changes. Sentinel-1 radar and a water-plant class became core parts of the plan, and RQ2 now tests SWOT's "suspect" readings.

Why these tanks? We checked all of Sri Lanka

6th of 187

Hurulu Wewa's rank among Sri Lanka's big lakes (1 km² or more), by best-quality SWOT readings

- All 5 of ours are in the island's top 22, and they are the top 5 in our study area
- The island's top tanks are spread from Puttalam to Hambantota, far from our village tanks
- Ours sit next to the 8 village tanks and share the same 2 SWOT passes
- All 5 have daily Irrigation Department gauges; 2 of the island's top 5 tanks do not



Best quality: quality flag 0, crossover flag 0–1, dark water 10% or less (Köhn and Nielsen, 2026).

Research questions

1

How accurately can deep learning map tank water from Sentinel-1 and Sentinel-2, even under cloud or water plants?

Measured by IoU and F1 against hand-labelled water masks

2

How accurate is SWOT on reservoirs and village tanks, and how many "suspect" readings can a learned screen keep?

Measured by error in cm against gauges, on reservoirs the screen never saw

3

Can a model trained on about 100 gauged reservoirs turn SWOT height and water area into % full, for tanks it never saw?

Measured by error in % of capacity, on reservoirs it never saw

4

Can a deep-learning model forecast each tank's storage 2–4 weeks ahead better than a "no change" forecast?

Measured by error as a % of capacity at 1, 2 and 4 weeks

Who does what: one model each

Three models in one chain: A finds where the water is, B how much there is, and C what comes next.

MEMBER A

AI water maps

RQ1

- Labelled water masks of Sri Lankan tanks
- A water-plant class for covered tanks
- Model: a Sentinel-1 and Sentinel-2 water map (U-Net)

Output: the water area of every tank

MEMBER B

SWOT and storage

RQ2 RQ3

- SWOT heights, checked against gauges
- A trust screen that cleans every reading
- Model: % full from height and area (gradient boosting)

Output: how full every tank is

MEMBER C

Forecast and system

RQ4

- Daily storage of 117 reservoirs, rain and evaporation
- The chain from picture to forecast, after every pass
- Model: a 2–4 week storage forecast (LSTM)

Output: the forecast and the dashboard

All three: field visits on SWOT pass dates, labelling water masks, the thesis and the research paper.

Member A: AI water maps

RQ1 How accurately can deep learning map tank water from Sentinel-1 and Sentinel-2, even under cloud or water plants?

Data analysis

- Label water masks of our tanks, with a "water under plants" class
- Score the maps: IoU and F1 against the labels
- Area error against Kanhindawewa's drone survey, by sensor, season and plant cover
- Compare with the simple rules and with SWOT's own area on the same days

Technical work

- U-Net (EfficientNet-B0) trained on S1S2-Water, then fine-tuned on our tanks
- Inputs: Sentinel-1 VV and VH, NISAR L-band radar, Sentinel-2, slope
- Pipeline: find each picture day, download it, map the water, measure the area

Output: water area for every tank on every picture day, with its accuracy

Member B: SWOT and storage

RQ2 RQ3

How accurate is SWOT, and can a model trained on about 100 reservoirs turn SWOT height and water area into % full for new tanks?

Data analysis

- SWOT minus gauge on every pass: the steady gap and the spread in cm
- Trust screen: how many "suspect" readings it keeps, and their error
- Training pairs: daily storage with SWOT height and water area, at about 100 reservoirs
- Storage error in % of capacity on reservoirs the model never saw, against one formula and Imbulana's equation

Technical work

- Hydrocron pipeline: each tank's lake ID, every pass
- Trust screen: gradient boosting on SWOT's quality fields, track distance, lake size and season
- Storage model: gradient boosting in which more height or area never means less water
- Inputs every tank has: height, area, outline, catchment and slope (Copernicus DEM)

Output: SWOT accuracy report, a screen that cleans every reading, and a model that says how full any tank is

Member C: Forecast and system

RQ4 Can a deep-learning model forecast each tank's storage 2–4 weeks ahead, from recent storage, rain and evaporation, better than "no change"?

Data analysis

- Seasonal average for each tank, from past years only
- Forecast error at 1, 2 and 4 weeks, against "no change" and the seasonal average
- Tested on years and reservoirs the model never saw
- The whole chain, picture to forecast, at the village tanks and on Kanhindawewa's drone survey

Technical work

- Forecast model: an LSTM with a gradient-boosting baseline, trained on 117 reservoirs' daily storage (2020–2026), rain and evaporation
- The chain after every pass: A's water map and SWOT's height go into B's model
- Dashboard: storage, safe limits and the forecast for each tank
- Automatic updates with GitHub Actions

Output: a 2–4 week forecast for every tank, the running chain and the dashboard

What we will deliver, and when

- A labelled dataset and an AI water map that works under cloud and water plants
- A SWOT trust screen, and a model that turns height and area into % full
- Storage every 5–12 days for all 13 tanks
- A 2–4 week storage forecast, shown on a dashboard
- A research paper, for example for UoM's ICITR conference

Nov 2026

Proposal, ethics and data requests

15 Dec 2026

Study area confirmed after the cascade and name checks

Dec 2026 – Apr 2027

Labelled data, gauge readings, and the three models: water map, storage model, forecast

Mar – May 2027

Storage estimates, validation and dashboard

30 Jun 2027

Thesis, paper and final submission

Datasets and links

Every dataset is free and public. Click a name to open it.

Water height (SWOT)

- **SWOT Lake and River Explorer** click any tank to see its readings
- **NASA Earthdata: SWOT lake product, version D** the files we use
- **Hydrocron** the free service that gives each lake's heights

Water area (pictures)

- **Copernicus Browser** Sentinel-1 and -2 pictures of our tanks
- **Planetary Computer: Sentinel-1 radar** terrain-corrected radar
- **ASF: NISAR radar** L-band radar
- **JAXA ALOS-2 radar on AWS** L-band radar
- **S1S2-Water** AI training data, 65 areas in 29 countries

Ground truth

- **Irrigation Department: reservoir water level** the daily sheet
- **Irrigation Department: reservoir status** web page
- **Irrigation Department: reservoir map layer** full supply levels
- **Kanhindawewa drone survey** area and capacity every 1 ft (PDF)
- **Mahaweli Authority: reservoir status reports** daily levels

Weather (for the forecast)

- **CHIRPS** daily rainfall
- **ERA5** weather, for evaporation

Our work

- **Project website: swot.mehara.work** every tab, with the team plan

Thank you

Questions and feedback are welcome.

Data sources

- SWOT: NASA/JPL and CNES, Lake Single-Pass product version D (PO.DAAC, Hydrocron)
- Sentinel-1 and Sentinel-2: ESA Copernicus (Earth Search and Microsoft Planetary Computer)
- Reservoir levels and outlines: Irrigation Department of Sri Lanka
- Tank outlines © OpenStreetMap contributors; Sri Lanka boundary: geoBoundaries
- SWOT illustration: NASA/JPL-Caltech

See SWOT readings for any lake:
SWOT Lake and River Explorer